STAT 109: Introductory Biostatistics

Summer Session 1, 2026 — Online (async)

Quick Start

- Start with the course path for the recommended flow through the material.
- See the syllabus for policies and grading.
- All readings are in lectures.
- Assessments lists projects, labs, and quizzes.
- Find software links and summaries in resources.
- Canvas is where you submit work, take quizzes, see grades, get feedback, and chat with classmates.

Instructor

Dr. Rosanna Overholser
Email: rho3@humboldt.edu

Welcome to STAT 109: Introductory Biostatistics!

This website contains Rosanna Overholser's materials for an asynchronous version of STAT 109: Introductory Biostatistics at Cal Poly Humboldt. An asynchronous course means you work through the materials on your own schedule, with support from an expert and your peers along the way. My goal for you isn't just to finish assignments. By the end of the course, you will have a portfolio of five projects and tworeference sheets that you can continue to use, build on, and share for years to come. And perhaps a few fellow Lumberjacks to meet up with when we all return to campus in the fall.

Where you are

Most of what you read and work through lives on this website: lectures, labs and project instructions. Canvas is where you submit work, take quizzes, see grades, get feedback, and chat with classmates. The course path lays out a recommended order through the the material in five weeks; the syllabus has the official policies and grading information for the course.

What you'll learn

In this class you'll learn statistical ideas that people use to make decisions and test claims with data. You'll learn the concepts, but you'll also learn by doing—carrying out analyses with technology.

When you're working with a real dataset, point-and-click tools only get you so far. Data usually needs wrangling and cleaning before you can run a test or fit a model. Programming lets you do that work efficiently and leaves a record of what you did. That record matters for reproducible research: someone else (including future you) can follow what happened to the data from start to finish. That's a big part of why we use code instead of only clicking through menus.

We'll use R, which is free and widely used by statisticians and researchers who work with messy data. Python is another popular option; its data-processing syntax is similar enough to R's that you'll be able to pick it up quickly if you need to later.

We run R in Google Colab notebooks, which are a common way to run Jupyter notebooks with no local installation needed.

What you'll build

By the end of the course you'll have created two reference guides: one for statistical concepts (definitions, notation, methods you want to keep organized) and one for the R programming language (functions, code patterns, notes from your notebooks).

You'll also complete five projects: applied analyses with write-ups and code. You'll update the reference sheets a little every module; each project module adds one new project to your portfolio. For three of the projects, you'll chose a research question that you are interested and then apply methods from class to answer it. For the four, you'll answer a classmate's question and for the fifth, you'll compete with your classmates to make the most accurate set of predictions.

My hope is that you'll use include these five projects in your portfolio to demonstrate your understanding of the material and your ability to apply it to solve problems that people care about.

Why is the course like this?

In 2022, Cal Poly Humboldt became California's third polytechnic university. Being a polytechnic means emphasizing learning by doing: developing knowledge and skills through hands-on projects connected to the places where we live, work, and play.

This course follows that philosophy. Rather than focusing primarily on watching lectures and taking exams in artificial circumstances, you will learn statistics by conducting investigations, analyzing data, building statistical models, and evaluating evidence. By the end of the course, you will have practiced the same kinds of activities that professionals in business, education, healthcare, government, and community organizations use to answer questions and make decisions.

At the same time, we continue Humboldt's tradition of place-based learning by connecting with one another and the communities and environments that shape our lives through informal discussions and the course projects.

Cal Poly Humboldt's motto, *Light and Truth*, reminds us that education is a search for understanding. Statistics is one of the tools we use in that search. By collecting evidence, asking thoughtful questions, and carefully evaluating uncertainty, we can develop a clearer understanding of the world around us.

This course also reflects values that are central to Cal Poly Humboldt: caring for the environment and for our community. Whether we are studying forests, wildlife, public health, or everyday decisions, statistics helps us learn from evidence and contribute thoughtfully to the well-being of people and the planet.

Finally, this course is built around a particular vision of teaching and learning. Rather than viewing the professor as a "sage on the stage" whose job is to deliver answers, I see learning as a shared journey of inquiry. Together we will ask questions, gather evidence, build models, test ideas, and learn from mistakes. Along the way, you will have access to many resources, including your classmates, course materials, modern computational tools, AI assistants, and me. The goal is not simply to find answers, but to develop the habits of mind needed to ask good questions, think critically about evidence, work effectively with others, and solve challenging problems. My role is not only to teach statistics, but also to guide and support you as you build those skills.